# Fake News Detection in the Age of LLMs: Methods, Datasets, and Challenges

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***Abstract*—The spread of fake news and disinformation on digital media has become a pressing problem with profound social, political, and economic ramifications. The latest break throughs in machine learning, deep learning, and large language models (LLMs) have given rise to innovative methods for the detection and prevention of false information spreading. This paper provides an exhaustive overview of the methods and datasets applied for detection of fake news, emphasizing the shift from early methods like support vector machines and TF-IDF embeddings to recent transformer-based and graph-based architectures. Particular focus is given to multilingual detection, low-resource environments, and the application of social context to improve detection performance. In addition, recent benchmark datasets and evaluation methods are examined to demonstrate emerging research directions. By combining these contributions, this review offers an organized insight into the strengths and weaknesses of current methods as well as directions for future research to construct reliable, interpretable, and scalable systems for detecting misinformation**

**Index Terms—*Fake News Detection, Large Language Models(LLMs), Social Media, Natural Language Processing(NLP)***

## I. INTRODUCTION

The explosive expansion of social media platforms has facilitated unprecedented information spreading but also augmented the contagion of misinformation and disinformation at an alarming rate. Misinformation and disinformation have the potential to alter public perception, sway elections, spark violence, and weaken institutions, and so their detection is a pressing research task in natural language processing (NLP), data mining, and social network analysis.

Early detection techniques utilized rule-based and statistical methods, but the magnitude and sophistication of disinformation have spurred the use of machine learning and deep learning methods. Conventional supervised models like Support Vector Machines (SVM) with TF-IDF representations worked well for text classification, whereas recent work utilizes more complex architectures like graph neural networks (GNNs), ensemble architectures, and hybrid models to capture semantic and context dependencies. Dual and hybrid systems based on deep learning, along with early detection mechanisms, have been explored extensively. Recent work also emphasizes the contribution of large language models (LLMs) to addressing multilingual and low-resource scenarios, yet generalization across datasets and languages poses continuing challenges. Furthermore, misinformation's user-centric nature makes it easier to be viral, and detection systems must consider social context and propagation processes. These advancements clearly reflect a move toward harmonizing accuracy, explainability, and scalability in reinforcing fake news detection on heterogeneous platforms.

This survey brings together methods, data, and assessment strategies from traditional to cutting-edge methodology. Gathering existing contributions gives a systematic overview of fake news detection studies, highlights enduring gaps, and proposes future research directions to construct strong, understandable, and scalable detection systems.

## II. LITERATURE SURVEY

The problem of detecting fabricated news has been studied broadly in recent years using a diverse range of methodologies, datasets, and testing approaches. This section gives an organized description of previously done work with a view to exploring methodological contributions and utilized datasets

### *A. Systematic Reviews and Surveys*

A number of works have aggregated existing methods and given an organized review of fabricated news detection. A systematic review of social network misinformation detection had over 670 articles with focus on datasets, evaluation metrics, and detection pipelines [1]. Another survey in early days of misinformation detection classified methods as content-based and propagation-based methods [2]. A wider methodological overview was given in [3], whereas [4] highlighted detection methods specific to social networks. Explainability and real time detection were recently identified as open issues [5], unified machine learning approaches such as deep, hybrid, and traditional ones [6], and established benchmark datasets for testing.

### *B. Classical Machine Learning Methods*

Traditional supervised learning techniques were used in a number of studies for fake news detection, typically on typical text-based datasets. Robust detection performance was attained with SVM and TF-IDF features in [7]. Machine learning techniques including Naive Bayes and Decision Trees were applied on publicly available news corpora [8]. Graph Neural Networks (GNNs) were employed to capture relational structure among users and posts [9]. These experiments proved that while traditional approaches are still effective, their scalability and adaptability are usually compromised when compared to deep learning.

**1) Deep Learning Based Methods:** Deep learning has emerged as a prominent approach to fake news detection. Ex tensive discussions of deep learning architectures like CNNs, RNNs, and hybrid models were presented in [10]. Double deep models merging textual and contextual features were proposed in [11]. Hybrid frameworks unifying machine learning with neural models were discussed in [12]. Identifying fake news on the Social Web of Things by using machine learning pipelines was investigated in [13]. These methods typically draw on the datasets like LIAR, FakeNewsNet, and social media corpora (Twitter, Facebook) that offer text as well as contextual metadata.

### *C. Multilingual and Low-Resource Language Efforts*

A crucial research interest lies in processing multilingual and low-resource languages. Fact-checking platforms for low resource Indian languages were discussed in [14]. Large language models were used to augment detection in low-resource environments, showcasing how they can generalize across languages [15]. These papers emphasize dataset diversity since the majority of current benchmarks are English-biased.

### *D. Rumor and Early Detection Models*

Rumor detection and early detection of fake news are essential subdomains. A survey paper on rumor and rumor-source detection methods discussed joint detection methods [16]. A theory-based model for early fake news detection targeted temporal patterns and propagation signals in social media datasets [17]. Such approaches emphasize the significance of timeliness in detection since misinformation spreads quickly in initial phases.

### *E. User-Centric and Social Perspectives*

Finally, user-centric research has questioned how misinformation is disseminated and how users perceive it. A mixed methods analysis investigated user-friendly features of misinformation on social media [18]. Some of them point out the role of psychological and interface-level factors and promote future detection systems to include social and behavioral data alongside textual features.

## III. CHALLENGES

Despite the major advances in misinformation detection, current methods still encounter a series of key challenges that hinder their efficacy and scalability.

### *A. Lack of Novelty and Redundancy*

A significant portion of the current literature is comprised of survey-based studies or incremental modifications of existing models. This has created redundancy and constrained methodological innovation, precluding the field from progressing toward more effective and practical solutions. Multiple frameworks repeat current techniques without solving real world misinformation ecosystems' complexity.

### *B. Scalability and Computational Complexity*

A number of proposed models, especially those that combine deep learning architectures, graph neural networks, or hybrid systems, show promising accuracy but at the cost of computational efficiency. Heavy resource usage limits their applicability in real-time, large-scale scenarios like mass events, elections, or emergencies where instant detection is crucial.

### *C. Dataset Limitations and Generalizability*

The performance of misinformation detection largely relies on the existence of high-quality datasets. Yet, most datasets have limited size, language, and domain scope, tending pre dominantly to English-language content. This narrows the generalizability of the model across varied linguistic, cultural, and contextual environments significantly. Moreover, most datasets become outdated very rapidly since misinformation campaigns are constantly evolving.

### *D. Semantic and Contextual Understanding*

Conventionally used techniques like Bag-of-Words and TF IDF identify surface-level lexical patterns but cannot represent higher-level semantic and pragmatic aspects of language. Though advanced models based on embeddings and trans formers provide better performance, they are still limited by addressing subtle misclassifications like satire, irony, implicit statements, and highly sophisticated AI-written materials. These shortfalls decrease the robustness of present systems in actual detection.

### *E. Early Detection Challenges*

Misinformation is especially difficult to detect at an early stage because there might not be adequate propagation signals available in the early stages of diffusion. Content-based methods, although theoretically good for early detection, tend to perform poorly in reality since misinformation at the point of creation might not always have clear lexical or semantic indicators. This makes systems less proactive to prevent mass damage.

### *F. Robustness and Cross-Domain Adaptation*

Dual and hybrid models have strong performance on certain datasets and domains but tend to lack generalizability to unseen data distributions. Their lack of robustness under adversarial manipulation and inability to be ported to alternative platforms, languages, and content modalities are a lasting bottleneck. Structured environments-trained models tend to underperform in noisy, real-world environments.

### *G. Ethical, Social, and Governance Issues*

The application of automated misinformation detection also presents urgent ethical problems. Models have the potential to amplify biases, stifle authentic discussion, and facilitate censorship. Excessive filtering of content has the risk of destroying public faith, and sub-optimal filtering can amplify misinformation-related harms. In addition, the absence of governance frameworks makes it difficult to hold systems accountable for fairness and transparency in their deployment

TABLE I
SUMMARY OF MISINFORMATION AND FAKE NEWS DETECTION PAPERS

| Sr. | Title | Author(s) | Year | Methodology | Pros | Cons |
|---|---|---|---|---|---|---|
| 1 | Misinformation Detection in Social Networks: A Systematic Literature Review | Zafer Duzen et al. | 2022 | Systematic Literature Review of 670+ articles, emphasis on sources, datasets, ML/AI approaches. | Summarizes large body of work; identifies trends and techniques. | No new method; purely synthesis. |
| 2 | A Survey on Rumor Detection and Ru- mor Source Detection Problems with Joint Detection Approaches | Otabek Sattarov et al. | 2024 | Literature review of rumor and source detection; classified via ML, DL, graph-based, probabilistic models. Comparative analysis by accuracy, recall, F1-score, source localization error. | Holistic approach—detects and traces source; bridges gap between problems. | High complexity; scalability and data reliability issues. |
| 3 | An Extensive Survey on Deep Learning-based Fake News Detection | M. F. Mridha et al. | 2021 | Combines influence maximization with ML. Uses LCLD, RMD for node detection, TextCNN for classification, cascaded model for spread prediction. Evaluated on BuzzFeed dataset. | Accurately identifies nodes; suppresses spread; validated on real data. | High computational complexity; relies on TextCNN; may remove useful nodes. |
| 4 | Multimodal Multilingual Caption-aware Fake News Detection (MMCFND) | Shubhi Bansal et al. | 2024 | Uses pre-trained encoders for text/image, generates captions, fuses features into classifier. New dataset MMIFND for Indic languages. | Effective for Indic languages; multi- modal approach; new dataset contribution. | No specific cons reported. |
| 5 | FactDRIL: Factorization of Fact-Checks for Low Resource Indian Languages | Shivangi Singhal et al. | 2021 | Created FactDRIL dataset across 11 Indian languages + English (22,435 samples). Annotated under multilingual, multimedia, multi-domain framework. | Fills dataset gap for regional languages; comprehensive design. | Challenges in consistency, translation, cultural nuances; low-resource limitations. |
| 6 | User-Friendly Aspects of Social Me- dia Misinformation: A Mixed Methods Analysis | Ilona Fridman et al. | 2025 | Structured review + dataset from cancer-related misinformation on X (Twitter). NLP + Lasso regression to identify linguistic cues. | Provides linguistic cues usable by non-experts; bridges algorithmic and user detection. | Accuracy (73%) limited; manual labeling bias; may miss subtle misinformation. |
| 7 | From Scarcity to Capability: Empow- ering Fake News Detection in Low- Resource Languages with LLMs | Hrithik Majumdar Shibu et al. | 2025 | Introduces BanFakeNews-2.0 dataset for Bangla. Uses transformer-based models with Q-LoRA fine-tuned LLMs. | Outperforms traditional models; dataset and model publicly released. | No cons mentioned. |
| 8 | Advanced Text Analytics – GNN for Fake News Detection in Social Media | Anantram Patel et al. | 2022 | ATA-GNN model on text only. Topic modeling + clustering + GNN integration to learn semantic structures. | Captures semantic relationships; works without propagation/user data; better than GNN baselines. | Depends on clustering quality; computationally heavy; cannot handle multi- modal content. |
| 9 | Fake News Suppression through the So- cial Web of Things | Nabamita Deb et al. | 2023 | Hybrid: Influence maximization (LCLD, RMD) + TextCNN for classification. Propagation model tested on BuzzFeed dataset. | Integrates graph + deep learning; sup- presses misinformation at source nodes; validated on real dataset. | Removing nodes may harm real info flow; computationally complex; effectiveness may decline in dynamic net- works. |
| 10 | A Survey on Analysis of Fake News Detection Techniques | Shailender Kumar et al. | 2021 | Survey of ML/DL approaches and feature extraction techniques for FND. | Comprehensive review; maps techniques and challenges. | No new method proposed. |
| 11 | Revealing the Truth: Fake News Detec- tion with SVM and TF-IDF | Randi Rizal et al. | 2025 | Used Kaggle datasets; preprocessing with TF-IDF; models: SVM, RF, Logistic Regression. SVM achieved 99.6% accuracy. | High accuracy (99.6%); TF-IDF effective; SVM outperformed others. | Limited datasets; ignores semantics/context; over-reliance on accuracy metric. |
| 12 | Machine Learning Based Fake News Detection | P. Santhiya et al. | 2023 | Fraud detection in job postings via TF-IDF, BoW. Models: SVM, RF, Naive Bayes; ensemble voting. RF best (98.18%). | High accuracy; ensemble increases reliability; diverse features captured. | Dataset update needed; BoW/TF-IDF lack deep semantics; ignores metadata signals. |
| 13 | An Analysis and Identification of Fake News using ML Techniques | Ashish et al. | 2024 | Majority voting hybrid of Gradient Boosting, XGBoost, MLP. Compared with 7 other ML models. | Hybrid model outperformed standalone; broad comparison across algorithms. | No specific cons reported. |
| 14 | Investigating the Potential of Dual Deep Learning for Fake News Detection | Hounaida Moalla et al. | 2024 | Proposed DuSTraMo dual-stream transformer model + ERAF-News dataset (4 categories: real, fake, AI- generated true, AI-generated fake). | Dual-stream better than single; ERAF-News valuable; detects AI-generated news. | Complexity higher; weaker performance on BART-generated news; generalization across languages needs care. |
| 15 | Early Detection of Fake News: A Theory-driven Model | Xinyi Zhou et al. | 2020 | Content-only model (lexicon, syntax, semantic, dis- course features) inspired by psychology theories. Used ML classifiers for early detection. | Enables early detection before social spread; interpretable features; out performs state-of-art in tests. | Some features less intuitive; discourse-level weak alone; single-model reliance limits robustness. |
| 16 | Overview of detecting fake news: From a new perspective | Bo Hu et al. | 2024 | Comprehensive survey analyzing the diffusion process of fake news to summarize three intrinsic characteristics (intentional creation, heteromorphic transmission, controversial reception); classifies and discusses existing detection approaches. | Provides a new perspective; thorough classification of approaches; reveals technological trends for future research. | Complexity in addressing fake news spread; findings may require updates with new developments. |
| 17 | A taxonomy of social network fake news detection | Upasna Sharma et al. | 2024 | Describes forms of fake news; presents taxonomy for detection; analyzes DL-based algorithms, monolingual/multilingual models, datasets, and mitigation tools. | Comprehensive taxonomy and analysis; addresses multilingual detection; lists useful datasets and tools. | Lack of robust multilingual databases; technical challenges in detection. |
| 18 | Fake news detection: new trends and challenges | Hemang Thakar et al. | 2024 | Explores advanced ML models, NLP techniques, and cross-modal analysis leveraging textual, visual, and contextual cues. | Highlights emerging trends; emphasizes interdisciplinary collaboration for com- prehensive solutions. | Adversarial attacks; data scarcity; do- main adaptation issues. |
| 19 | From Misinformation to Insight: Ma- chine Learning Strategies for Fake News Detection | Despoina Mouratidis et al. | 2025 | Develops ML framework integrating NLP and DL; evaluates traditional ML and advanced DL models with vectorization techniques across multiple datasets. | Superior performance with BERT models; robust metrics (MCC, ROC-AUC); addresses ethical implications. | Ethical concerns, including algorithmic bias and censorship. |
| 20 | From Misinformation to Insight: Ma- chine Learning Strategies for Fake News Detection | Despoina Mouratidis et al. | 2025 | Develops ML framework integrating NLP and DL; evaluates traditional ML and advanced DL models with vectorization techniques across multiple datasets. | Superior performance with BERT models; robust metrics (MCC, ROC-AUC); addresses ethical implications. | Ethical concerns, including algorithmic bias and censorship. |
| 20 | A Hybrid Attention Framework for Fake News Detection with Large Language Models | Xu et al. | 2025 | Integrates textual statistical features with LLM generated deep semantic features via a hybrid attention mechanism; uses attention heatmaps and SHAP for model interpretability. Evaluated on WELFake dataset. | F1-score improvement (+1.5%); scalable and interpretable framework. | Adds interpretability complexity; needs evaluation across diverse datasets. |

## IV. FUTURE DIRECTIVES

Future research will need to navigate both technical challenges and general societal issues by following the directions below:

### *A. Multilingual and Cross-Cultural Detection:*

It is essential to develop models and data that are sensitive to language differences and cultural contexts. Future research needs to emphasize multilingual corpora, cross-lingual transfer learning, and culture adaptation of detection systems to guarantee fair and consistent performance in global settings.

### *B. Efficient and Scalable Architectures*

Light and computationally inexpensive models are required for real-time detection in high-volume settings. Methods like model compression, knowledge distillation, and lightweight transformer architectures can tradeoff accuracy with scalability to facilitate practical deployment at scale.

### *C. Context-Aware and Explainable Systems*

Future systems will need to draw on discourse-level, pragmatic, and socio-contextual cues beyond surface textual characteristics. Multimodal data integration, such as images, video, and network dynamics, can add richer context to classification. Prioritizing explainability will also promote user trust, regulatory acceptability, and system decision interpretability.

### *D. Adversarial Robustness and Security*

Improving models to resist adversarial attacks and evasive deception strategies is an essential area of research. This involves the creation of adversarial training methods, resilient evaluation tasks, and systems that can detect manipulated or AI-generated content intended to mislead detectors.

### *E. Low-Resource and Under-Researched Domains*

Research needs to extend to low-resource languages, minority groups, and under-represented areas where disinformation has similarly dire effects but fewer technical protections. Transfer learning, synthetic data creation, and multilingual large language models can assist in bridging this resource imbalance.

### *F. Integration with Social Network Dynamics*

Merging content-based methods with network-driven analysis provides a promising avenue. Simulating the spread of misinformation, the identification of influential users, and diffusion patterns can make it possible to detect more precisely and suppress more effectively dangerous stories.

### *G. Ethics, Fairness, and Collaborative Governance*

Lastly, ethical considerations should be at the heart of future models. Frameworks should be developed so that they reduce bias, do not censor, and ensure fairness among different demographic groups. Multistakeholder partnerships between researchers, social media companies, policymakers, and civil society must come together to build governance structures that promote accountability as well as sustainable deployment of misinformation detection systems.

## CONCLUSION

The exponential spread of misinformation and disinformation over internet channels is an enduring problem for researchers and practitioners. This survey has comprehensively evaluated classical, deep learning, and combined methods and recent developments that involve large language models (LLMs). Although machine learning classics like SVMs and TF-IDF continue to be useful for structured text classification, they cannot provide the semantic depth needed to detect more subtle misinformation. Deep learning models, hybrid models, and graph-based approaches are better in performance but suffer from scalability and generalizability challenges. One vital finding from the literature reviewed here is the significance of low-resource and multilingual language detection, in light of the English-centric nature of datasets. FactDRIL and BanFakeNews-2.0 are among the recent datasets indicating a push to bridge this gap, and multimodal and attention-based models are promising to detect richer contextual features. Early detection, adversarial robustness, ethical deployment, and computational efficiency remain problems to tackle. Directions for future research need to focus on efficient, explainable, and domain-independent adaptable systems that co-integrate content and network dynamics. Building multilingual benchmarks, real-time lightweight architectures, and governance structures that balance fairness and transparency will be critical. Relying ultimately on not just technical ingenuity but also interdisciplinary co-operation to protect information integrity in increasingly sophisticated digital environments will be the hallmark of effective misinformation detection.

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